GridShift

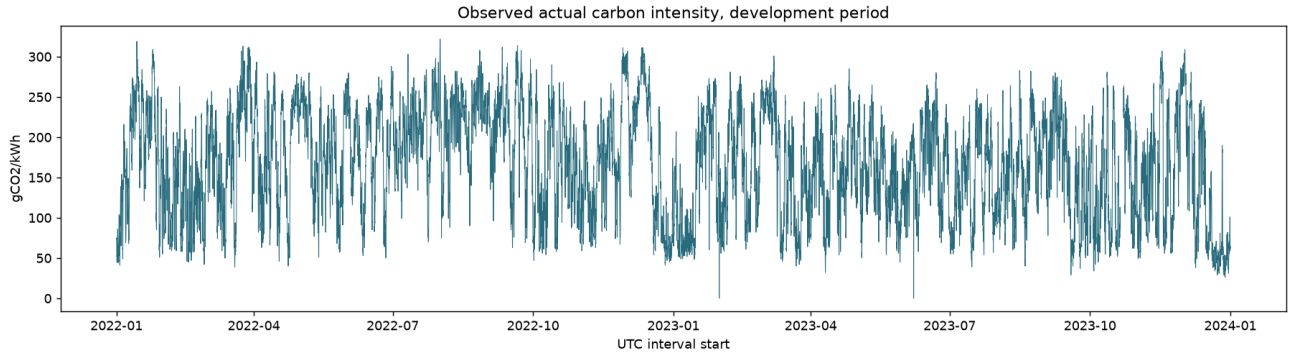
Carbon-aware GPU workload scheduling with classical time-series forecasts

Research report generated from verified local artifacts

Study ID: d2dee2d358735d7d

Target: NESO estimated GB carbon intensity in gCO2/kWh

Scheduling emissions are modeled from declared energy assumptions.



Abstract

This report evaluates a frozen classical forecasting and scheduling study. It separates validation selection from locked-test measurement.

Modeled isolated-job benefit under declared assumptions. The certified seven-day paired block estimate is 12.414% (95% interval 10.951 to 13.828%) across 2913 scheduled primary-policy jobs; the bootstrap paired 2913 jobs. This estimates modeled operational CO2 under the declared job, energy, and deadline assumptions.

The source series contains NESO estimated actual carbon intensity. The scheduling outcomes are modeled estimates under stated job and energy assumptions.

Research question and scope

The study asks whether half-hourly forecasts of NESO estimated GB carbon intensity can improve feasible GPU workload placement against earliest-feasible execution and simple seasonal controls.

The primary experiment uses isolated counterfactual jobs. A secondary one-GPU queue replay tests a simple reservation policy under synthetic arrivals.

The carbon oracle observes realized future intensity. It is an unattainable opportunity bound, not a candidate for deployment.

component	definition
Target	NESO estimated actual, gCO2/kWh
Forecast path	Half-hourly, configured horizons
Primary contrast	Validation-selected schedule versus earliest feasible
Queue	One GPU, synthetic arrivals, non-preemptive reservations

Data and provenance

The data manifest records normalized hash `f57548d5bc197f884b5b35bb84faaaadcb38b7b46dc3e8c925a150b8ea5a5a1e`.

The configured source is `https://api.carbonintensity.org.uk`. The target spans `2022-01-01T00:00:00Z` through `2026-01-01T00:00:00Z`.

The provider's historical forecast values do not have verified issue-time provenance in the inspected interfaces. They are not treated as fair as-issued competitors. [S1-S5]

Data-quality audit

The table reproduces the stored quality audit. It reports coverage and grid checks without converting the planned thresholds into measured facts.

The raw reconstruction verified 209 source chunks and 70128 UTC half-hour slots. It identifies 122 intervals without a raw API record and 56 records with an explicit null actual. These categories come from the raw interval audit; diagnostics.input_absent_intervals describes the development-series input and cannot establish raw API absence.

measure	value
slots	70128
observed	69950
missing	178
overall_coverage	0.9975
minimum_monthly_coverage	0.9388
minimum_test_monthly_coverage	1
unique_utc_intervals	true
regular_grid	true
data_gate_pass	true
availability_is_assumed	true

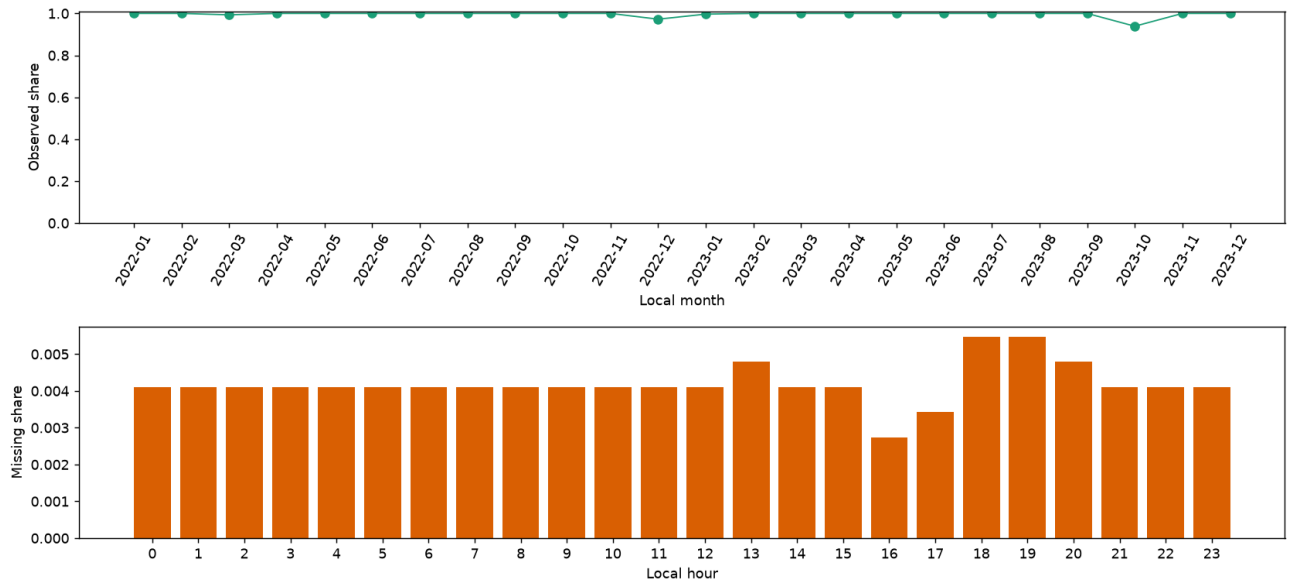


Figure. Development-period missingness by month and local hour. Source: artifacts/diagnostics/figures/missingness.png.

Development series

This plot uses development observations only. It informs transformations and candidate construction. It is not a test-year result.

The source target remains separate from provider forecasts. Missing target values are not filled from the forecast field. [S2, S4]

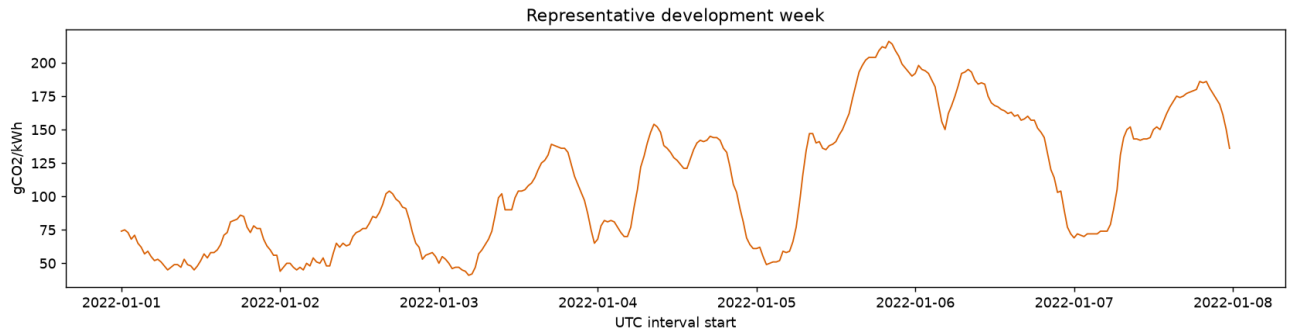


Figure. Representative development week from the stored diagnostics.

Seasonality and decomposition

The diagnostic record reports daily ACF at lag 48 as `0.52289` and weekly ACF at lag 336 as `0.26307` when those values exist.

Robust STL at period 48 is descriptive. Its centered endpoint behavior cannot support a real-time claim. Residual whiteness does not establish forecast accuracy. [S9-S13]

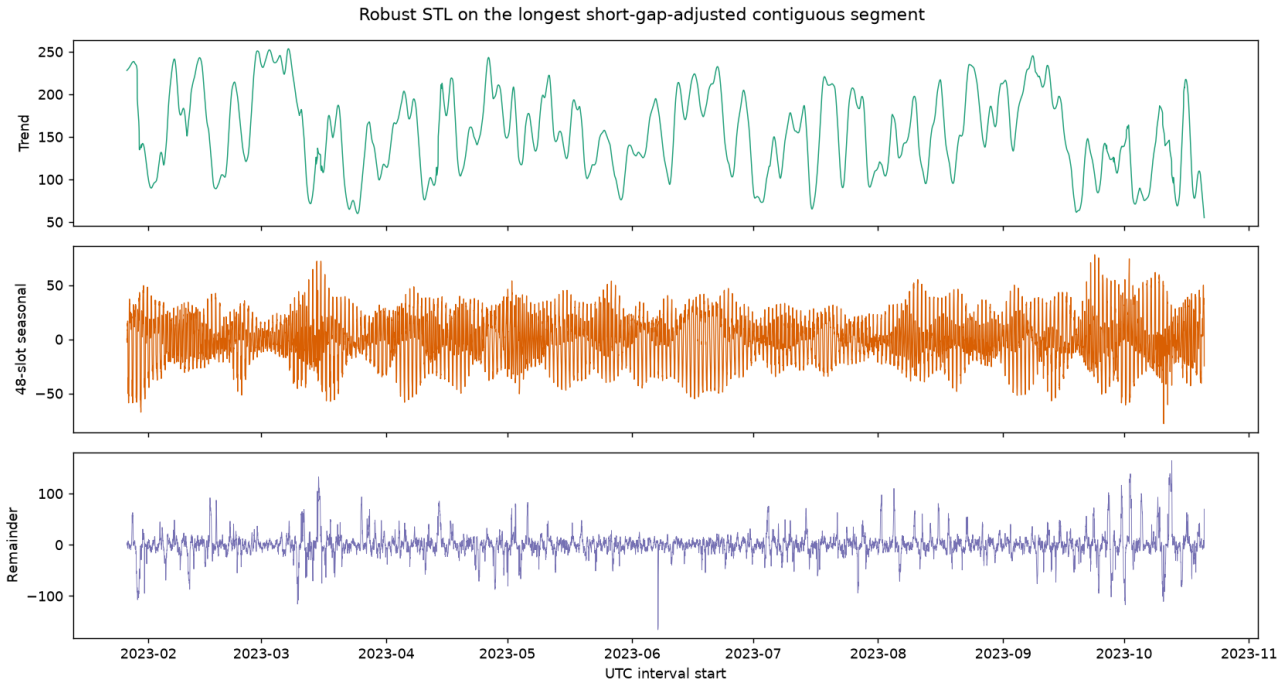


Figure. Robust STL decomposition on the recorded development segment.

Stationarity diagnostics

ADF tests address a unit-root null under specified deterministic terms. KPSS uses a different null. Conflicting or inconclusive tests do not select a forecasting model by themselves. [S11]

The diagnostics record contains 20 ADF rows and 8 KPSS rows. Missing rows remain absent rather than being filled with assumed p-values.

Recorded level ADF p-values range from 6.769e-21 to 3.5813e-12. Level KPSS p-values are 0.08477, 0.08268. KPSS values at 0.1 or 0.01 are lookup-table bounds, not exact tail probabilities. Read each against its specified regression, sample period, and lag cap. These tests do not establish stability across years or prove white noise.

variant	regression	autolag	status	sample_n	selected_lag	test_statistic	p_value	critical_5pct
levels_identity	c	AIC	ok	12828	48	-11.42	6.769e-21	-2.862
levels_identity	c	BIC	ok	12828	48	-11.42	6.769e-21	-2.862
levels_identity	ct	AIC	ok	12828	48	-11.51	2.595e-18	-3.411
levels_identity	ct	BIC	ok	12828	48	-11.51	2.595e-18	-3.411
levels_log1p	c	AIC	ok	12828	48	-8.5	1.248e-13	-2.862
levels_log1p	c	BIC	ok	12828	48	-8.5	1.248e-13	-2.862
levels_log1p	ct	AIC	ok	12828	48	-8.564	3.581e-12	-3.411
levels_log1p	ct	BIC	ok	12828	48	-8.564	3.581e-12	-3.411
first_difference_identity	c	AIC	ok	12827	48	-14.04	3.31e-26	-2.862
first_difference_identity	c	BIC	ok	12827	48	-14.04	3.31e-26	-2.862
first_difference_log1p	c	AIC	ok	12827	48	-18.49	2.125e-30	-2.862
first_difference_log1p	c	BIC	ok	12827	48	-18.49	2.125e-30	-2.862

Forecast model catalogue

The study uses AR, MA, ARMA, ARIMA, and daily-seasonal SARIMA structures. Baselines include persistence, daily and weekly seasonal-naive, and past-only calendar mean.

AR, MA, and ARMA can be nested ARIMA cases. Effective duplicates are counted once in comparisons. Unsuccessful fits remain part of the run ledger. [S9, S10, S12]

AR uses past levels; MA uses past innovations rather than a rolling arithmetic average. ARMA combines those terms. ARIMA applies (1-B)^d before stationary ARMA dynamics. SARIMA adds seasonal polynomials and (1-B^48)^D.

family	role
AR	Stationary autoregression
MA	Past innovation dependence
ARMA	Stationary mixed dynamics
ARIMA	Integrated level model
SARIMA	Daily seasonal state-space model
Simple baselines	Persistence, seasonal-naive, calendar mean

Lag selection and information criteria

AIC = -2 log L + 2k; BIC = -2 log L + k log(n_effective). AICc adds 2k(k+1)/(n_effective-k-1). k includes variance and deterministic parameters.

These representative first-anchor shortlist rows belong to separate family, transformation, and differencing groups. Their absolute IC values must not be pooled into a leaderboard.

All attempted orders and failures remain in selection_ledger.csv. Aggregate within-anchor AICc and BIC ranks shortlist orders; chronological validation chooses representations.

family	transform	order	aic	bic	aicc	converged_anchors
AR	identity	[7, 0, 0]	5.4e+04	5.407e+04	5.4e+04	12
MA	identity	[0, 0, 8]	5.823e+04	5.83e+04	5.823e+04	12
ARMA	identity	[3, 0, 2]	5.403e+04	5.408e+04	5.403e+04	12
ARIMA	identity	[3, 1, 3]	5.398e+04	5.403e+04	5.398e+04	12
SARIMA	identity	[1, 0, 1]	5.462e+04	5.465e+04	5.462e+04	12
AR	log1p	[8, 0, 0]	-2.614e+04	-2.607e+04	-2.614e+04	12
MA	log1p	[0, 0, 8]	-2.222e+04	-2.215e+04	-2.222e+04	12
ARMA	log1p	[3, 0, 1]	-2.612e+04	-2.607e+04	-2.612e+04	12
ARIMA	log1p	[2, 1, 1]	-2.601e+04	-2.598e+04	-2.601e+04	12
SARIMA	log1p	[1, 0, 1]	-2.572e+04	-2.569e+04	-2.572e+04	10

Representative validation residual diagnostics

Ljung-Box at lag 48 checks remaining serial dependence with model degrees of freedom. All lags, innovation normality, and ACF/PACF plots remain in validation/residuals/.

model_id	n_contiguous	lb_stat	lb_pvalue
AR(7,0,0)	8226	1544	2.513e-297
MA(0,0,8)[log1p]	8226	1.273e+04	0
ARMA(3,0,2)	8226	1203	6.107e-224
ARIMA(3,1,3)	8226	1173	1.653e-218
SARIMA(1,0,1)x(1,0,0,48)	8226	2459	0

Small p-values indicate remaining serial dependence. The innovation-distribution table accompanies the report. Printed zero p-values represent numerical underflow. These diagnostics and imperfect predictive coverage prevent a calibrated Gaussian-uncertainty claim.

Chronological evaluation design

Forecasts use eligible history, persist before scoring, and are compared on common target masks. The test year remains outside validation selection.

Report MAE, RMSE, mean error, daily-seasonal MASE, predictive interval coverage, interval width, and sample counts by lead time. Avoid MAPE when targets may be zero. [S9-S13]

setting	value
Validation start	2024-01-01T00:00:00Z
Validation end	2025-01-01T00:00:00Z
Test start	2025-01-01T00:00:00Z
Test end	2026-01-01T00:00:00Z
Reported horizons	[1, 4, 12, 24, 48]
Forecast path slots	48

Validation forecast outcomes

The validation-selected forecast champion is `SARIMA(1,0,1)x(1,0,0,48)`.

The strongest simple validation baseline is `persistence`. These selections precede locked-test scoring.

These are common-origin scores. End-to-end scores and all required families remain in the accompanying Markdown report and source tables.

model_id	group	horizon	n	mae	rmse	mean_error	mase	coverage80	coverage95
SARIMA(1,0,1)x(1,0,0,48)	all	1	15683	7.124	9.864	-0.02633	0.1736	0.8325	0.9371
SARIMA(1,0,1)x(1,0,0,48)	all	4	15680	15.05	20.58	-0.05673	0.3671	0.7511	0.8854
SARIMA(1,0,1)x(1,0,0,48)	all	12	15672	27.92	36	-0.1264	0.6808	0.6681	0.8422
SARIMA(1,0,1)x(1,0,0,48)	all	24	15660	33.82	43.57	-0.2568	0.8226	0.6982	0.872
SARIMA(1,0,1)x(1,0,0,48)	all	48	15654	40.21	51.23	-0.3095	0.9783	0.7198	0.8885
persistence	all	1	15683	8.781	12.45	-0.02002	0.2146	0.9996	1
persistence	all	4	15680	18.99	26.11	-0.03705	0.4647	0.9758	0.9986
persistence	all	12	15672	35.44	45.41	-0.08518	0.868	0.8613	0.9731
persistence	all	24	15660	39.12	50.36	-0.2317	0.9565	0.8341	0.9545
persistence	all	48	15654	41.43	54.26	-0.4704	1.01	0.8085	0.9367

Locked-test forecast outcomes

These rows report the recorded locked-test results for the validation-selected champion and strongest simple baseline.

The exporter does not choose a new winner from test outcomes. A post-test revision requires another untouched evaluation period.

model_id	group	horizon	n	mae	rmse	mean_error	mase	coverage80	coverage95
SARIMA(1,0,1)x(1,0,0,48)	all	1	16465	6.926	9.362	-0.02102	0.1766	0.814	0.9309
SARIMA(1,0,1)x(1,0,0,48)	all	4	16465	14.88	20.08	-0.05526	0.3788	0.726	0.8695
SARIMA(1,0,1)x(1,0,0,48)	all	12	16465	27.79	35.85	-0.1243	0.7074	0.637	0.8212
SARIMA(1,0,1)x(1,0,0,48)	all	24	16465	33.41	43.22	-0.1281	0.8524	0.6895	0.8508
SARIMA(1,0,1)x(1,0,0,48)	all	48	16465	39.26	50.73	0.1809	1.002	0.7063	0.87
persistence	all	1	16465	9.101	12.57	-0.04422	0.2313	0.9999	1
persistence	all	4	16465	20.07	27.17	-0.1076	0.5095	0.9719	0.9991
persistence	all	12	16465	37.39	47.63	-0.2743	0.9491	0.8256	0.9636
persistence	all	24	16465	39.25	50.55	-0.4479	1	0.8039	0.9446
persistence	all	48	16465	39.8	51.8	-0.4904	1.017	0.8012	0.9288

Paired forecast-loss comparison

Positive loss reduction favors the validation-selected classical model over the validation-selected simple forecast. The equal-horizon common-origin contrast is secondary and descriptive.

block_days	paired_origin_count	estimate_loss_reduction_pct	ci95_loss_reduction_pct_low	ci95_loss_reduction_pct_high
7	16465	16.03	13.37	18.7

Cumulative forecast-path errors

Path MAE and RMSE score every half-hour in each prefix in gCO2/kWh. Prefix-mean MAE scores the error of the time-average intensity.

Only complete observed prefixes are scored. Common-origin masks exclude any competitor fallback. These descriptive results do not change endpoint-based selection.

model_id	horizon	n_complete_paths	path_mae	path_rmse	prefix_mean_mae
SARIMA(1,0,1)x(1,0,0, 48)	1	16465	6.926	9.362	6.926
SARIMA(1,0,1)x(1,0,0, 48)	4	16465	11	15.42	10.63
SARIMA(1,0,1)x(1,0,0, 48)	12	16465	19.04	26.45	17.59
SARIMA(1,0,1)x(1,0,0, 48)	24	16465	25.14	34.08	21.71
SARIMA(1,0,1)x(1,0,0, 48)	48	16465	31.43	42.08	25.54
persistence	1	16465	9.101	12.57	9.101
persistence	4	16465	14.69	20.81	14.38
persistence	12	16465	25.83	35.65	23.99
persistence	24	16465	32.23	43.08	27.37
persistence	48	16465	37.54	49.57	30.31

Scheduling formulation

For a candidate start, the scheduler sums forecast intensity times interval-overlap energy across the full job duration. It obeys hard deadlines and capacity. Exact cost ties favor the earliest start.

Primary releases are independent counterfactual cases. They are not a simultaneous workload trace. The one-GPU queue simulation uses synthetic arrivals and immutable reservations. [S5]

assumption	value
Default duration (hours)	2
Default slack (hours)	10
Declared power (kW)	0.4
PUE	1.2
Primary selected model	calendar_mean

Primary scheduling outcomes

Modeled isolated-job benefit under declared assumptions. The certified seven-day paired block estimate is 12.414% (95% interval 10.951 to 13.828%) across 2913 scheduled primary-policy jobs; the bootstrap paired 2913 jobs. This estimates modeled operational CO2 under the declared job, energy, and deadline assumptions.

A simple-baseline deployment choice does not establish added scheduling value from classical forecasting. Persistence's flat path resolves to earliest execution. Read all policy rows before attributing the benefit to model complexity.

policy	model_id	n_scheduled	mean_realized_grams	paired_savings_pct	oracle_regret_grams
earliest	NA	2913	124	NA	26.81
seeded_random	NA	2913	123.5	0.4067	26.3
oracle	oracle	2913	97.22	21.61	0
forecast	AR(7,0,0)	2913	114.1	8.017	16.87
forecast	ARIMA(3,1,3)	2913	114.8	7.477	17.53
forecast	ARMA(3,0,2)	2913	113.9	8.198	16.64
forecast	MA(0,0,8)[log1p]	2913	118.3	4.639	21.05
forecast	SARIMA(1,0,1)x(1,0,0,48)	2913	105.7	14.81	8.436
forecast	calendar_mean	2913	108.6	12.41	11.41
forecast	daily_naive	2913	107.7	13.14	10.51
forecast	persistence	2913	124	0	26.81
forecast	weekly_naive	2913	107.6	13.25	10.37

Primary uncertainty

The paired moving-block bootstrap keeps neighboring cases together. The seven-day estimate is primary when present; three- and fourteen-day estimates are sensitivity checks.

A non-positive interval or a null estimate remains a valid negative result. A positive estimate does not establish measured or causal grid emissions.

block_days	paired_job_count	estimate_savings_pct	ci95_savings_pct_low	ci95_savings_pct_high	is_primary_block
3	2913	12.41	11.1	13.65	False
7	2913	12.41	10.95	13.83	True
14	2913	12.41	10.64	14.07	False

Selected policy versus the strongest simple forecast

No classical deployment was selected; selected-policy improvement is simple timing value, not classical added value

paired_job_count	estimate_savings_pct	ci95_savings_pct_low	ci95_savings_pct_high
2913	12.41	10.95	13.83

All declared duration/slack scenarios are reproduced in the complete scenario table in gridshift_report.md and the primary_summary.csv source. They remain secondary descriptive comparisons.

Monthly scheduling outcomes

Monthly figures show the default duration and slack where the source table records both. They describe temporal variation and do not replace the prespecified aggregate contrast.

month_utc	scenario_id	policy	model_id	n_eligible_jobs	n_scheduled	mean_realized_grams	paired_savings_pct	fallback_rate
2025-01	duration_2h_slack_10h	earliest	NA	248	248	161.5	NA	0
2025-01	duration_2h_slack_10h	seeded_random	NA	248	248	162.2	-0.4032	0
2025-01	duration_2h_slack_10h	oracle	oracle	248	248	135.3	16.23	0
2025-01	duration_2h_slack_10h	forecast	AR(7,0,0)	248	248	151.4	6.252	0
2025-01	duration_2h_slack_10h	forecast	ARIMA(3,1,3)	248	248	153.1	5.203	0
2025-01	duration_2h_slack_10h	forecast	ARMA(3,0,2)	248	248	151.7	6.106	0
2025-01	duration_2h_slack_10h	forecast	MA(0,0,8)[log1p]	248	248	157.5	2.487	0
2025-01	duration_2h_slack_10h	forecast	SARIMA(1,0,1)x(1,0,0,48)	248	248	142.8	11.62	0
2025-01	duration_2h_slack_10h	forecast	calendar_mean	248	248	140.1	13.27	0
2025-01	duration_2h_slack_10h	forecast	daily_naive	248	248	145.6	9.862	0
2025-01	duration_2h_slack_10h	forecast	persistence	248	248	161.5	0	0
2025-01	duration_2h_slack_10h	forecast	weekly_naive	248	248	149.1	7.679	0
2025-02	duration_2h_slack_10h	earliest	NA	224	224	141.8	NA	0
2025-02	duration_2h_slack_10h	seeded_random	NA	224	224	140.1	1.171	0

Queue replay outcomes

Queue emissions must be read with offered jobs, completed jobs, rejections, deadline misses, and unmatched work. A lower total caused by rejected work is not a carbon saving.

The queue summary comes from the `test` artifact path.

offered_load	seed	slack_hours	policy	offered_jobs	completed_jobs	rejected_jobs	matched_savings_pct	interpretation
0.3	1729	6	earliest	1411	1404	7	0	descriptive completion intersection; service differs
0.3	1729	6	selected	1411	1380	31	7.322	descriptive completion intersection; service differs
0.3	1729	6	strongest_simple	1411	1404	7	0	descriptive completion intersection; service differs
0.3	1729	12	earliest	1411	1410	1	0	descriptive completion intersection; service differs
0.3	1729	12	selected	1411	1404	7	9.207	descriptive completion intersection; service differs
0.3	1729	12	strongest_simple	1411	1410	1	0	descriptive completion intersection; service differs
0.3	2718	6	earliest	1324	1317	7	0	descriptive completion intersection; service differs
0.3	2718	6	selected	1324	1294	30	5.703	descriptive completion intersection; service differs
0.3	2718	6	strongest_simple	1324	1317	7	0	descriptive completion intersection; service differs
0.3	2718	12	earliest	1324	1324	0	0	descriptive completion intersection; service differs
0.3	2718	12	selected	1324	1318	6	8.291	descriptive completion intersection; service differs
0.3	2718	12	strongest_simple	1324	1324	0	0	descriptive completion intersection; service differs

Bounded sensitivity checks

Six forecasting variants use four three-day quarterly episodes in validation year 2024. These checks do not establish full-year robustness.

Constant power and PUE scale absolute costs algebraically. They cannot establish robustness to measured energy variation.

month	variant	n_scheduled	paired_savings_pct
1	default	17	18.07
1	delay_0min	17	18
1	delay_60min	17	18.14
1	window_90days	17	18.08
1	window_365days	17	18.07
1	native_missing	17	18.07
4	default	17	2.424
4	delay_0min	17	2.694
4	delay_60min	17	3.246
4	window_90days	17	2.424
4	window_365days	17	5.926
4	native_missing	17	2.424

Deployment and demonstration

The study records `calendar_mean` as the validation-selected scheduling model.

This export does not claim a live GPU scheduler, Slurm integration, cloud workload execution, or public deployment. A local dashboard demo must use the saved versioned artifacts and preserve stale, missing, and infeasible states.

At an illustrative supply intensity of 100 gCO2/kWh, gross savings permit 0.15397 kWh of overhead per isolated counterfactual case. Computing energy is unmeasured, so no net-emissions claim follows.

Limitations

The NESO series is estimated generation-related CO2. It is not a complete lifecycle or all-GHG inventory. [S2, S5]

Historical publication latency and revisions remain uncertain. Without verified vintages, retrospective replay is pseudo-real-time. [S2, S4]

Power, PUE, duration, and synthetic arrival assumptions do not establish measured GPU behavior.

Retrospective NESO estimated actuals have no verified historical publication vintages.

Thirty-minute publication delay is an assumed replay policy.

Power, PUE, known durations, and arrivals are synthetic. No GPU energy was measured.

Costs are modeled intensity-weighted operational CO2, not actual avoided grid emissions.

National GB intensity is a proxy for a hypothetical GB-powered workload.

Predictive intervals condition on fitted parameters and may be poorly calibrated.

Queue completion intersections are descriptive when policies accept different jobs.

Net computing overhead, marginal emissions, and embodied emissions are unmeasured.

The untouched test covers 2025 only. No post-test model promotion is justified.

Local packaging and runtime verification do not establish public deployment.

Modeled isolated-job benefit under declared assumptions. The certified seven-day paired block estimate is 12.414% (95% interval 10.951 to 13.828%) across 2913 scheduled primary-policy jobs; the bootstrap paired 2913 jobs. This estimates modeled operational CO2 under the declared job, energy, and deadline assumptions.

Reproducibility and evidence map

Results manifest SHA-256: 8cd196dbbe13512f1e09f2c07e285ce4f476bae673184ab8cd56b92fae58d743.

Data hash: f57548d5bc197f884b5b35bb84faaaadcb38b7b46dc3e8c925a150b8ea5a5a1e.

Configuration digest: 6f0831112aaed65643e9f88b6d31ac5d31222897d4484c461d9103ac94810ccb.

Protocol hash: b6876f7c8f058f9242eb5852a580384bfa0a1365c76f69a509fa975f8377c68c.

Software source hash: 16d9038d1d153c6d68293c5cc8ef0a902381366d5b8ecd993c9cccf03477fda0.

Study ID: d2dee2d358735d7d.

claim	evidence
Validation selection	results.json selections and the recorded order-search ledger when present
Locked-test forecast result	test/forecast_metrics.csv
Primary scheduling estimate	C:\Users\viraj\Documents\Codex\2026-10-22\plan-gridshift-a-complete-ats-mini\outputs\GridShift\artifacts\runs\d2dee2d358735d7d\test\single_job\primary_summary.csv
Paired uncertainty	C:\Users\viraj\Documents\Codex\2026-10-22\plan-gridshift-a-complete-ats-mini\outputs\GridShift\artifacts\runs\d2dee2d358735d7d\test\single_job\primary_bootstrap.csv
Queue service and emissions	C:\Users\viraj\Documents\Codex\2026-10-22\plan-gridshift-a-complete-ats-mini\outputs\GridShift\artifacts\runs\d2dee2d358735d7d\test\queue\queue_summary.csv

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